No Spoiler Please! A TinyBERT Model to Detect Spoiler Reviews on IMDb

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Abstract

This report contains the explanation of the work done for the final project of the Natural Language Processing course. The professor in charge of the course is Professor Alfio Ferrara, University of Milan. The project focused on the application of knowledge distillation, a technique used to compress large language models into smaller and more efficient versions. This is achived by training a smaller model (the student) to replicate the behavior of a larger model (the teacher). In particular, this project had the aim to implement a spoiler detector for IMDb's movie reviews: a BERT-base model has been used as the teacher model, while a TinyBERT_General_4L_312D has been used as the student model. The experiments shows that TO BE EXPENDED

1 Introduction

Very often, before even seeing a film, we decide to read some reviews because we're curious to know the opinions of those who have already seen it. In such a situation, it's not uncommon to encounter reviews that, whether intended to spoil something or not, include various types of spoilers (e.g., "The movie is great, but at the end, character X dies/does this or that"). Clearly, we want to avoid this, but platforms like IMDb don't warn us in advance about such reviews.

IMDb is by far the most popular global source for movies and TV shows, and it also includes user-submitted reviews. But, as mentioned above, there's no spoiler alert for reviews. What if there were a browser extension (running locally on users' PCs) that retrieves the text of a review and tells us "Spoiler Alert!" before we can even start reading it?

With the recent evolution of LLMs, this task can be accomplished by fine-tuning (adding a binary classification layer on top of the model) a sufficiently large pre-trained model, such as BERT.

Language model pre-training, such as BERT, has significantly improved the perfomances of many NLP tasks. However, pre-trained language models are usually computationally expensive, so it is difficult to efficiently execute them on resource-restricted devices (such as the user's PC). A novel Transformer distillation method, specially designed for knowledge distillation of the Transformer-based models, allows to effectively transfer to a smaller student model called TinyBERT the plenty of knowledge from a large pre-trained BERT model. A two-stage learning framework for TinyBERT which performs Transformer distillation at both the pretraining and task-specific learning stages ensures that TinyBERT can capture the general-domain as well as the task-specific knowledge in BERT. [1]

2 Related Work

So far, three main studies have defined the paradigm of model compression via knowledge transfer. The first, by Hinton et al. [2], introduced the seminal concept of Knowledge Distillation (KD), focusing on transferring the so-called dark knowledge from a large Teacher to a compact Student. Their approach minimizes the distance between the output probability distributions of the two networks, softened by a temperature parameter T. The second and more specialized study, by Jiao et al. [1], proposes a variant named TinyBERT, specifically designed for Transformer-based architectures. While Hinton's method operates on the final output layer, Jiao et al. argue that for complex language models, this is insufficient. Their method leverages a layer-to-layer distillation strategy, forcing the student to mimic not only the teacher's prediction but also its intermediate representations, such as attention matrices and hidden states. The third, by Ji & Zhu [3], attempts to provide a mathematical justification for why knowledge distillation works, demonstrating that the use of hard labels (ground truth) are necessary during distillation to counteract the so-called "imperfect teacher" (i.e., a teacher who makes incorrect predictions). This way, the student can understand how much he can trust the teacher's predictions.

2.1 Aim of the work

2.2 Data

2.2.1 Preprocessing

2.3 Implementation

2.3.1 Structure of the models

The choosen teacher model was BERT-base-uncased [4], while the choosen student model was TinyBERT_General_4L_312D [1]: this because this is exactly the TinyBERT model from Jiao et al with General Knowledge Distillation already performed (and therefore not necessary for my work) and, more importantly, it allows to implement the same techniques they use to handle the mismatch in the number of layers and dimensions of hidden states and embeddings. Table 1 shows the differences in the number of parameters between the two models.

Table 1 Parameters of the teacher and student models.

Model	Layers	Attention Heads	Embeddings	Hidden States	Total Parameters
BERT-base-uncased	12	12	768	768	110M
TinyBERT_General_4L_312D	4	12	312	312	14.5M

2.3.2 BERT Fine-tuning

2.3.3 Task Specific Distillation

3 Results

4 Conclusion

References

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